

Results

Fixture Release Packet I

The fixture release packet contains fourteen segments spanning four classification levels: UNCLASSIFIED (ten segments), CUI (one segment), SECRET (two segments), and TOP_SECRET (one segment). Four segments carry source controls (drawn from the HUMINT, SIGINT, and IMINT disciplines). Twenty-two redaction decisions are applied across those four segments, using all five bounded reasons: `source_identity`, `operational_detail`, `time_place_selector`, `legal_privilege`, and `privacy`.

Fixture Release Packet I

The audit produces:

Fixture Release Packet I

- ▶ **Releasable:** true (no error-level findings after redaction).
- ▶ **Release safety score:** weighted by error count, warning count, and mosaic risk.
- ▶ **Redaction coverage:** 1.0 (all sensitive segments have at least one decision).
- ▶ **Mosaic risk score:** residual markers normalized by segment and pattern count.
- ▶ **Findings:** warning-level findings for residual markers in sanitized text.

The canonical run's measured values are bound to `output/reports/redaction_audit.json` by the manuscript-binding tests:

Fixture Release Packet I

Metric	Measured value
Segment count	14
Redaction decision count	22
Reviewer record count	3
Release safety score	0.798
Redaction coverage	1.0
Mosaic risk score	0.012
Findings	2 warning-level
Releasable	true
Final release recommended	false
Approvals	3

Fixture Release Packet I

Because the intelligence policy blocks on warnings and the fixture retains two warning-level residual markers, `final_release_recommended` is false even though the packet is releasable and the review gate is approved.

Source-Safe Redaction Ledger I

The redaction ledger records each decision with:

Source-Safe Redaction Ledger I

- ▶ A decision ID derived from the SHA-256 of the segment ID, span, reason, and replacement.
- ▶ The segment ID, start, end, span length, reason, and replacement.
- ▶ A `valid_span` flag indicating whether the span falls within the segment text.
- ▶ A `source_span_sha256` hash of the redacted text—present only for valid spans, absent for invalid or orphan decisions.

Source-Safe Redaction Ledger I

The ledger never exposes source text. It provides reproducible audit evidence without disclosure risk.

The normal analysis stage writes this evidence to `output/data/release_ledger.json`. Its companion `output/reports/redaction_audit.json` contains policy outcomes, review-gate status, findings, scores, counts, and paragraph metadata. Both canonical files omit narrative text, review rationales, and reviewer identities; the comprehensive sanitized packet is evaluated in memory rather than serialized to the public output tree.

Segment Hash Manifest I

The hash manifest records, for each segment:

Segment Hash Manifest I

- ▶ `source_sha256`: SHA-256 of the original segment text.
- ▶ `public_sha256`: SHA-256 of the redacted segment text.

Segment Hash Manifest I

This allows downstream verification that the public release matches the audited version without comparing source text directly.

Review Gate I

The release gate requires three reviewer roles: originator, classification reviewer, and release authority. Each reviewer provides a non-empty rationale. The fixture reviews all approve, yielding:

Review Gate I

- ▶ **Approved:** true.
- ▶ **Approval count:** 3.
- ▶ **Required roles present:** classification_reviewer, originator, release_authority.
- ▶ **Findings:** none.

Review Gate I

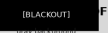
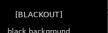
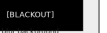
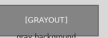
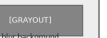
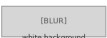
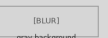
The `final_release_recommended` flag is true only when the packet is releasable, the review gate is approved, and no blocking warnings remain.

Visual Proof Matrix I

The development proof matrix produces sixteen base PDFs (four redaction styles $\times$ four backgrounds) and, when steganography is enabled, sixteen companion steganography PDFs with hash manifests. Each variant records:

Visual Proof Matrix I

- ▶ Base PDF filename, byte size, and SHA-256.
- ▶ Steganography PDF filename, byte size, and SHA-256.
- ▶ Hash manifest filename and SHA-256.
- ▶ Kmyth sidecar count and filenames (when Kmyth is available).

	White	Gray	Black	Blur
Blackout	 [BLACKOUT] white background	 [BLACKOUT] gray background	 [BLACKOUT] black background	 [BLACKOUT] blur background
Whiteout	 [WHITEOUT] white background	 [WHITEOUT] gray background	 [WHITEOUT] black background	 [WHITEOUT] blur background
Grayout	 [GRAYOUT] white background	 [GRAYOUT] gray background	 [GRAYOUT] black background	 [GRAYOUT] blur background
Blur	 [BLUR] white background	 [BLUR] gray background	 [BLUR] black background	 [BLUR] blur background

All sixteen variants apply the identical source-safe redaction decisions;
only the visual token and page background differ.

Styles: Blackout: [BLACKOUT] | Whiteout: [WHITEOUT] | Grayout: [GRAYOUT] | Blur: [BLUR]

Visual Proof Matrix II

Figure 1: Development proof matrix: four redaction styles (blackout, whiteout, grayout, blur) rendered across four PDF backgrounds (white, gray, black, blur); every one of the sixteen cells applies the identical source-safe redaction decisions.

Visual Proof Matrix I

fig. 1 renders the sixteen style-background combinations; only the visual token and the page background differ between cells.

Kmyth TPM Sidecar Production I

When Kmyth tools are runnable and a TPM backend is available, each variant produces two `.ski` sidecars:

Kmyth TPM Sidecar Production I

1. `{variant_id}.hashes.json.ski` — the hash manifest sealed against the TPM storage hierarchy.
2. `{variant_id}_steganography.pdf.ski` — the steganography PDF sealed against the TPM storage hierarchy.

The `.ski` files are ASCII-armored and contain PCR selections, policy or-values, the storage key public area, and the sealed data object. Unsealing requires the same TPM and platform configuration.

In the verified run, all sixteen variants produced both sidecars, yielding thirty-two `.ski` files total. The `kmyth-seal` binary's `FlushContext` patch ensured that consecutive seal invocations did not exhaust the `swtpm` transient object slots.

Residual Risk Detection I

The residual-risk detector scans sanitized text for common public-release leaks:

Residual Risk Detection I

Pattern	Regex
Email address	\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b
IPv4 address	\b(?:\d{1,3}\.){3}\d{1,3}\b
Coordinate pair	\b-?\d{1,2}\.\d{3,},\s*-\d{1,3}\.\d{3,}\b
Controlled dissemination	\b(?:NOFORN\ ORCON\ REL\ s+TO)\b
Collection discipline	\b(?:HUMINT\ SIGINT\ IMINT\ MASINT\ OSINT)\b
Compartment marker	\b(?:SCI\ TS_SCI\ TOP\ s+SECRET)\b
ISO calendar date	\b20\d{2}-\d{2}-\d{2}\b
Sensitive markers	HUMINT, SIGINT, source, selector, location

Residual Risk Detection I

Each detected marker generates a warning finding. The mosaic risk score aggregates residual markers across all segments.

Discussion

The exemplar demonstrates that disclosure control can be decomposed into orthogonal concerns: text-level audit, visual presentation, steganographic provenance, and hardware-backed sealing. Each concern is independently configurable and verifiable.

The visual proof matrix confirms that blackout, whiteout, grayout, and blur treatments produce equivalent source-safe outputs—only the visual token differs. The steganography layer adds provenance without altering the redaction decisions. Kmyth TPM sealing adds a hardware binding that ensures sealed sidecars can only be unsealed on the same platform configuration.

The mssim-to-swtpm protocol proxy is a necessary bridge on macOS, where no hardware TPM exists. The proxy handles three protocol mismatches: (1) platform commands use different command numbers, (2) the data channel wraps TPM commands in a 9-byte mssim header, and (3) swtpm's transient object slots are limited and must be flushed between seal invocations.

The FlushContext patch to kmyth-seal is a critical fix for batch sealing workflows. Without it, the second kmyth-seal invocation fails with “out of memory for object contexts” because the storage key from the first invocation remains loaded in the TPM's transient object table.

Limitations I

Limitations I

- ▶ The fixture data is invented; real release packets may require additional classification taxonomies and review-role policies.
- ▶ The swtpm software TPM does not provide hardware-level tamper resistance; production deployments should use a hardware TPM.
- ▶ The mssim proxy adds latency to each TPM command; batch sealing of thirty-two sidecars takes approximately thirty seconds.
- ▶ PDF password encryption is optional and uses AES-256; the password is not stored in the variant matrix.